

Estimation (Part 4)

STAT 4530/6530
Statistical Inference for Data Scientists

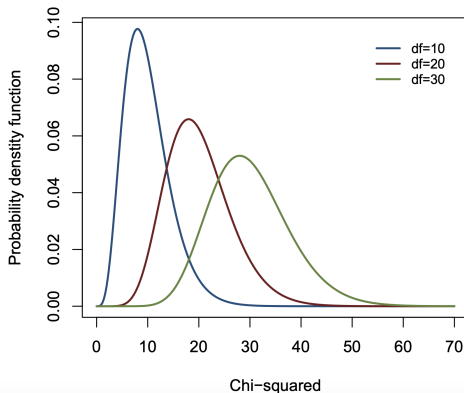
Construction of t distribution

We have touched on some important aspects of the t distribution. We will now discuss how to construct a random variable that has a t distribution.

The **chi-squared distribution with d degrees of freedom**, denoted by χ_d^2 , is the distribution of $Z_1^2 + \cdots + Z_d^2$ for d independent standard normal random variables $Z_1, \dots, Z_d$.

A χ_d^2 distribution is the special case of the gamma distribution with shape parameter $k = \frac{d}{2}$ and rate parameter $\lambda = \frac{1}{2}$.

Construction of t distribution



If $X \sim \chi_d^2$, then $E(X) = d$, $\text{Var}(X) = 2d$, and X has skewness coefficient $\sqrt{8/d}$.

As d increases, the mean and variance increase and the pdf becomes more bell-shaped. A normal approximation is reasonably good when d is at least about 30, although a slight skew still exists.

Construction of t distribution

A random variable with a t distribution can be constructed from random variables with standard normal and chi-squared distributions.

Let $Z \sim N(0, 1)$ and $X \sim \chi_d^2$, where Z and X are independent. Then the random variable

$$T = \frac{Z}{\sqrt{X/d}}$$

has a t distribution with d degrees for freedom.

Note that X is equivalent in distribution to $\frac{1}{d} \sum_{i=1}^d Z_i^2$, where the Z_i are standard normal random variables.

By the law of large numbers $\frac{1}{d} \sum_{i=1}^d Z_i^2$ converges in probability to $E(Z_i^2) = \text{Var}(Z_i) = 1$ as $d \rightarrow \infty$. It follows that for large d , T has an approximate standard normal distribution.

Pivotal Quantity Distribution

This construction of a random variable with a t distribution puts us in position to characterize the distribution of the pivotal quantity

$$T = \frac{\bar{Y} - \mu}{S/\sqrt{n}}$$

that is used to generate a confidence interval for μ .

For now, we identify the key steps required to do this and defer the details until later.

Pivotal Quantity Distribution

Step 1: If $Y_1, \dots, Y_n$ are independent observations from a $N(\mu, \sigma^2)$ distribution, then

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{\sigma^2} \sim \chi_{n-1}^2.$$

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Step 2: The pivotal quantity T can be expressed as

$$T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} = \frac{\sqrt{n}(\bar{Y} - \mu)/\sigma}{\sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}}},$$

where $\sqrt{n}(\bar{Y} - \mu)/\sigma$ is a random variable with a $N(0, 1)$ distribution, and $\sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}}$ is the square root of a χ_{n-1}^2 -distributed random variable divided by its degrees of freedom.

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Step 3: If $\sqrt{n}(\bar{Y} - \mu)/\sigma$ and $\frac{(n-1)S^2}{\sigma^2}$ are independent, then it will follow that T has a t distribution with $df = n - 1$. The required independence follows from the fact that $\bar{Y}$ and S^2 are independent random variables when constructed from independent random variables $Y_1, \dots, Y_n$ that follow a $N(\mu, \sigma^2)$ distribution.

Cauchy Distribution

The t distribution with 1 degree of freedom is known as the **Cauchy distribution**. Its pdf is

$$f(y) = \frac{1}{\pi(1 + y^2)}.$$

The quartiles of the Cauchy distribution are -1 and 1, but the distribution has very thick tails that die out very slowly as $|y|$ increases. The first and second moments (and therefore the mean and variance) are undefined and do not exist.

The Cauchy distribution is unusual in that many phenomena that otherwise hold quite generally do not apply to it. For instance, the central limit theorem does not apply. In fact, $\bar{Y}$ turns out to have a Cauchy distribution, so $\bar{Y}$ does not even converge in probability to 0 (the mean of a t distribution with $df > 1$).

Cauchy Distribution

Because of its thick tails, the Cauchy distribution is often used to model situations that are unusual in some way.

- In financial applications for instance, when a simulation uses a statistic that should have approximately a standard normal distribution under certain conditions, a Cauchy random variable is sometimes substituted to study the impact of a potential extreme event (often referred to as a *black swan*).

Comparing Two Population Means or Proportions

Most studies, rather than merely estimating a single mean or proportion, aim to compare two or more groups.

To compare means μ_1 and μ_2 , we treat $\mu_1 - \mu_2$ as a parameter and estimate it by the observed difference of sample means, $\bar{y}_1 - \bar{y}_2$.

The sampling distribution of $\bar{Y}_1 - \bar{Y}_2$ has expected value $\mu_1 - \mu_2$.

We will focus now on constructing a confidence interval for $\mu_1 - \mu_2$.

A Model for Comparing Means

Let $y_{i1}, \dots, y_{n_1,1}$ denote n_1 observations in the first sample and let $y_{i2}, \dots, y_{n_2,2}$ denote n_2 observations in the second sample.

Let s_1 and s_2 denote the sample standard deviations, which estimate the corresponding population standard deviations σ_1 and σ_2 .

To compare means, we introduce a statistical model as an approximation for the true relationship between them. Let Y_1 denote a randomly selected observation from group 1 and Y_2 a randomly selected observation from group 2. Our model assumes normality with common variability:

$$Y_1 \sim N(\mu_1, \sigma^2)$$

$$Y_2 \sim N(\mu_2, \sigma^2),$$

where the values of the parameters are unknown.

We can derive sampling distributions and resulting inferences under this assumed model structure.

A Model for Comparing Means

Such a model is a convenient simplification of reality — population distributions will not be exactly normal with identical variances.

Such model assumptions may seem worrisome, but the confidence interval we will present for $\mu_1 - \mu_2$ is robust against violations of this assumption, especially when $n_1 = n_2$ or they are close.

When the assumption of a common variance seems to be badly violated, we'll see that an alternative version of the method is available that does not require it.

When adequate, however, simpler models have benefits: they generalize directly to settings having additional variables. and inferences can be more powerful and simpler to summarize because of having fewer parameters than under more complex frameworks for the population structure.

A Model for Comparing Means

For this model, we estimate the common standard deviation σ by

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}} = \sqrt{\frac{\sum_{i=1}^{n_1} (y_{i1} - \bar{y}_1)^2 + \sum_{i=1}^{n_2} (y_{i2} - \bar{y}_2)^2}{n_1 + n_2 - 2}}$$

The pooled estimated s combines information from the two samples to provide a single estimate of variability.

Note that the term inside the square root is a weighted average of the two sample variances, with weight proportional to sample size.

A Model for Comparing Means

For the common-variance model,

$$\text{Var}(\bar{Y}_1 - \bar{Y}_2) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_2) = \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}.$$

With s as the estimate of σ , the estimated standard error of $\bar{Y}_1 - \bar{Y}_2$ simplifies to

$$se = \sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}} = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

To construct a confidence interval, we use the pivotal quantity

$$T = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where S is the random variable version of the pooled estimate s .

A Model for Comparing Means

The pivotal quantity T has a t distribution with $n_1 + n_2 - 2$ degrees of freedom. This leads to the following $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$:

$$(\bar{y}_1 - \bar{y}_2) \pm t_{\alpha/2, n_1+n_2-2} * s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

If the data show evidence of greatly different standard deviations for the two samples (with, say, one being at least 50% larger than the other one), it is better to use an alternative formula with:

- standard error replaced by $\sqrt{s_1^2/n_1 + s_2^2/n_2}$
- df replaced by a rather messy formula designed so that the corresponding pivotal quantity has distribution closely approximated by the t distribution.

The results are similar for the two approaches if s_1 and s_2 are close.

Example: Anorexia

Recall the example about cognitive behavioral therapy to treat teenage girls suffering from anorexia. This study, like most studies, had a control group that received no treatment. The results are summarized in the table below:

Group	Sample Size	Mean	Standard Deviation
Treatment	29	3.01	7.31
Control	26	-0.45	7.99

- The measured outcome was the weight change from the beginning of the study to the end of the study
- The researchers analyzed how the mean weight change compared for the treatment and control groups
- The girls in the study were randomly assigned to the cognitive behavioral therapy (Group 1) or to the control group (Group 2)

Example: Anorexia

Group	Sample Size	Mean	Standard Deviation
Treatment	29	3.01	7.31
Control	26	-0.45	7.99

The pooled estimate of the assumed common standard deviation is

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}} = \sqrt{\frac{(29 - 1) * (7.31)^2 + (26 - 1)(7.99)^2}{(29 - 1) + (26 - 1)}} = 7.64.$$

Also,

$$\bar{y}_1 - \bar{y}_2 = 3.01 - (-0.45) = 3.46$$

and

$$se = s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 7.64 \sqrt{\frac{1}{29} + \frac{1}{26}} = 2.06.$$

Example: Anorexia

Because $df = n_1 + n_2 - 2 = 53$, a 95% confidence interval for $\mu_1 - \mu_2$ is:

$$(\bar{y}_1 - \bar{y}_2) \pm t_{0.025, 53} * se = 3.46 \pm 2.006 * 2.06,$$

which gives us the interval $(-0.68, 7.59)$.

We interpret that the mean weight change for the cognitive behavioral therapy could be as much as 0.68 pounds lower or as much as 7.59 pounds higher than the mean weight change for the control group.

Because the confidence interval contains 0, it is plausible that the population means are identical. The confidence interval is relatively wide because the sample sizes are not large.

When a confidence interval for $\mu_1 - \mu_2$ contains only positive values, we can infer that $\mu_1 > \mu_2$. One that contains only negative values suggests that $\mu_1 < \mu_2$. Of course, labeling of groups is arbitrary, as is whether we estimate $\mu_1 - \mu_2$ or $\mu_2 - \mu_1$.

Confidence Interval Comparing Two Proportions

If the variable of interest in the populations is binary, the sample means $\bar{y}_1$ and $\bar{y}_2$ correspond to proportions. We are interested in comparing the true population proportions for the variable, p_1 and p_2 .

For independent samples of respective sizes n_1 and n_2 ,

$$\text{Var}(\bar{y}_1 - \bar{y}_2) = \text{Var}(\bar{y}_1) + \text{Var}(\bar{y}_2) = \frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}.$$

We want to construct a confidence interval for the difference $p_1 - p_2$ using the pivotal quantity

$$Z = \frac{(\bar{y}_1 - \bar{y}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

However, we do not know p_1 and p_2 , so we cannot plug them into the denominator to compute the standard error.

Confidence Interval for Comparing Two Proportions

One solution is to replace p_1 and p_2 in the aforementioned denominator (the standard error formula) by their maximum likelihood estimates subject to the constraint that $p_1 - p_2 = \Delta$, which we denote by $\hat{p}_1(\Delta)$ and $\hat{p}_2(\Delta)$. Then we use the following pivotal quantity:

$$Z = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta}{\sqrt{\frac{\hat{p}_1(\Delta)(1-\hat{p}_1(\Delta))}{n_1} + \frac{\hat{p}_2(\Delta)(1-\hat{p}_2(\Delta))}{n_2}}}$$

- For each possible value of Δ , the standard error uses the p_1 and $p_2 = p_1 - \Delta$ values that are maximum likelihood estimates for that particular Δ .
- Software implementations of this method analyzes this for all possible Δ , and the **score confidence interval** is the set of Δ values that yield $|z| \leq z_{\alpha/2}$.

Confidence Interval for Comparing Two Proportions

A simpler solution is to replace p_1 and p_2 in the standard error formula by their sample proportions $\bar{y}_1$ and $\bar{y}_2$. This yields the **Wald confidence interval**:

$$(\bar{y}_1 - \bar{y}_2) \pm z_{\alpha/2} \sqrt{\frac{\bar{y}_1(1 - \bar{y}_1)}{n_1} + \frac{\bar{y}_2(1 - \bar{y}_2)}{n_2}}$$

- However, as in the single-sample case, when the sample sizes are small and the true proportions are near 0 or 1, the score confidence interval has better performance.
- For example, the 95% score confidence interval tends to have actual coverage probability closer to 0.95 than does the 95% Wald confidence interval.

Example

A study used patients at six U.S. hospitals who were to receive coronary artery bypass graft surgery.

- The patients were randomly assigned to two groups.
- For one group, volunteers were instructed to pray for a successful surgery with a quick, healthy recovery and no complications. The prayer started the night before the surgery and continued for two weeks.
- The other group did not have volunteers pray for them.
- The measured response was whether medical complications occurred within 30 days of surgery.

Group	Complications	No Complications	Total
Prayer	315	289	604
No Prayer	304	293	597

Example

Let p_1 denote the probability of complications for those patients who have a prayer group, and let p_2 denote the probability of complications for the patients who do not have a prayer group.

The sample proportions are $\bar{y}_1 = 315/604 = 0.522$ and $\bar{y}_2 = 304/597 = 0.509$.

We can use R to construct 95% confidence intervals for $p_1 - p_2$:

```
> prop.test(c(315,304), c(604, 597), conf.level=0.95, correct=FALSE)
95 percent confidence interval:
-0.04421536  0.06883625    # Wald 95% confidence interval for pi_1 - pi_2
sample estimates:
   prop 1    prop 2 
0.5215232 0.5092127 

> library(PropCIs)           # uses binomial success count and n for each group
> diffscoreci(315, 604, 304, 597, conf.level=0.95)
95 percent confidence interval:
-0.04418699  0.06873177    # score 95% confidence interval for pi_1 - pi_2
```

The two confidence intervals are nearly identical and both contain 0, indicating that the probability of complications is similar for the two groups and may be equal.

The Bootstrap

Some statistics are too complex to have a simple standard error formula.

Even if the statistic has an approximately normal sampling distribution, without the standard error we cannot use the confidence interval formula of a point estimate plus and minus a quantile-score multiple of an estimated standard error.

A simple and powerful alternative is a computational resampling method known as the **bootstrap**.

The Bootstrap

The bootstrap treats the sample distribution as if it were the true population distribution and approximates features of the unknown sampling distribution by simulation.

- At each step the method samples n observations from the original data set of n observations, with replacement (so the same data point can appear multiple times).
- Each of the original n data points has probability $1/n$ of selection for each “new” observation.
- The new sample of size n has its own estimate of the parameter of interest θ .
- The bootstrap method repeats this sampling process a very large number of times, for instance, selecting 10,000 separate samples of size n and computing corresponding estimates $\{\hat{\theta}^{(1)}, \hat{\theta}^{(2)}, \dots, \hat{\theta}^{(10,000)}\}$. The empirical distribution of the $\hat{\theta}^{(b)}$ is called the **bootstrap distribution**.

The Bootstrap

The bootstrap distribution has shape and variability similar to the actual sampling distribution of $\hat{\theta}$ around θ , but its values vary around $\hat{\theta}$ rather than around θ .

- In other words, the sampling distribution of $\hat{\theta}^{(b)} - \hat{\theta}$ is similar to the sampling distribution of $\hat{\theta} - \theta$.
- This means that the standard deviation of the $\hat{\theta}^{(b)}$ values from the bootstrap distribution estimates a standard error for the point estimate $\hat{\theta}$ from the observed data.
- Also, the interval of values between the 2.5 and 97.5 percentiles of the simulated $\hat{\theta}^{(b)}$ values is a 95% confidence interval for θ .

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.